

How the Trade Filter Works

A step-by-step example: turning a long market string into a single trade signal.

Legend: blue underline = the pattern we found | **red bold** = the digit captured (the one right AFTER the pattern)

1. Setup and inputs

Raw string	110110000100011101011010
Filter 1 (F1) pattern	10
Filter 2 (F2) pattern	01

2. First layer — Filter 1

We scan the raw string for the pattern **10**. Every time we find it, we **capture the very next digit** immediately after it (shown in **red**). Then we resume scanning after that captured digit.

1 1 0 **1** 1 0 **0** 0 0 1 0 **0** 0 1 1 1 0 **1** 0 1 1 0 **1** 0

#	"10" found at	digit captured (next position)	value
1	positions 2-3	position 4	1
2	positions 5-6	position 7	0
3	positions 10-11	position 12	0
4	positions 16-17	position 18	1
5	positions 21-22	position 23	1

Combine the captured digits left to right → **Filter 1 output: 10011**

3. Second layer — Filter 2

Now we take the Filter 1 output **10011** and scan *it* for the pattern **01**. Same rule: capture the digit right after each match.

1 0 0 1 **1**

There is one 01 (at positions 3-4); the digit after it is **1**. **Filter 2 output (real trade signal): 1**

4. Trace it back to the market

That final **1** is the 5th digit of the Filter 1 output, which was captured at **position 23** of the original raw string. So out of the whole string, this is the single bit we trade:

1 1 0 1 1 0 0 0 0 1 0 0 0 1 1 1 0 1 0 1 1 0 **1**
0

(every other bit is greyed out — only position 23 becomes the real trade)

Summary

Even though the starting raw string is very long, running it through a two-layer filter distills all that data down to **just one final bit to trade**.

In the market: there are many signals available every session. These filters quietly record all of them, but we only actually **take a trade when the final filtered condition is met** — turning a flood of data into a few high-quality decisions.

Note: this example uses “scan-and-skip” matching (after capturing a digit, we resume scanning after it). If an engine instead lets a captured digit start the next match (“overlapping”), the Filter 1 output for this same string would be 100111.